EverVest

Building the Data Foundation for Evidence-Based Long-Term Retail Investing

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1. The funds and the backtest design

To evaluate different portfolio construction approaches for long-term retail investors, we constructed three investment universes: Equity, Cryptocurrency, and a Combined portfolio containing both asset classes. Annualisation was performed using 252 trading days for the Equity and Combined universes, and 365 calendar days for the Cryptocurrency universe, reflecting their respective trading calendars. Within each universe, five portfolio optimisation methods were implemented, namely **Equal Weight, Minimum Variance, Maximum Sharpe, Risk Parity, and an innovative approach, Minimum CVaR**, resulting in a total of fifteen funds. All portfolios are constrained to be **long-only and fully invested**, ensuring that portfolio weights remain non-negative and sum to one at every rebalance.

Portfolio optimisation methods

Methods	Equation
Equal Weight	$w_i = \frac{1}{N}, \quad i = 1, \dots, N$
Minimum Variance	$\min_{\mathbf{w}} \ \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}$
Maximum Sharpe	$\max_{\mathbf{w}} \frac{\mathbf{w}^\top \boldsymbol{\mu} - r_f}{\sqrt{\mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}}}$
Risk Parity	$\min_{\mathbf{w}} \frac{1}{2} \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w} - \frac{1}{N} \sum_{i=1}^{N} \ln(w_i)$, then $\mathbf{w} \leftarrow \frac{\mathbf{w}}{\sum_i w_i}$
Minimum CvaR	$\min_{\mathbf{w}} \ CVaR_{95\%}(-\mathbf{w}^\top \mathbf{R})$. *(Full CvaR equation see Appendix A)*

Notes: $\sum_{i=1}^{N} w_i = 1$, $w_i \geq 0$, ***(All portfolios are fully invested and long-only)***

The portfolio strategies were evaluated using an **out-of-sample backtest** to simulate how each fund would have performed under realistic investment conditions. An **initial estimation window** from **1 January 2020 to 31 December 2022** was used to estimate the portfolio weights, while the **2023 period** served as the out-of-sample testing window. To avoid look-ahead bias, portfolio weights for each rebalance were calculated using only information that would have been available at that point in time.

An expanding-window approach was adopted, where the estimation window increased each month as new market data became available. Portfolios were retrained and rebalanced on the first trading day of each month, with portfolio weights held constant until the next rebalance. This approach reflects how long-term investors periodically update their allocations while preserving all historical information accumulated throughout the sample period.

Expanding-window backtest:

$$\mathbf{w}_t = f(\{\mathbf{r}_s : s < t\}), \qquad r_{p,t} = \mathbf{w}_t^\top \mathbf{r}_t$$

To ensure a fair comparison across all portfolio strategies, a **risk-free rate of 0%** was assumed when constructing the Maximum Sharpe portfolios and calculating Sharpe ratios. In addition, a **transaction cost of 5 basis points (0.05%) per rebalance** was incorporated as a simplified representation of trading frictions, including bid-ask spreads and execution costs. The same transaction-cost assumption was applied consistently across all portfolio strategies so that differences in performance reflected differences in portfolio turnover rather than inconsistent cost assumptions across strategies.

2. Out-of-sample results and fund fact sheets

Portfolio performance was evaluated using annualised return, annualised volatility, Sharpe ratio, maximum drawdown, cumulative growth of $1 invested, and portfolio turnover, allowing both return and risk characteristics to be compared across optimisation methods. All reported results are based on the 2023 out-of-sample period using the walk-forward backtest described in Section 1. Unless otherwise stated, the performance metrics reported are net of the assumed 5 basis point transaction cost per rebalance.

Key performance metrics (see ***Appendix A*** for further details):

Performance Metrics	Equation
Growth of $1	$W_T = \prod_{t=1}^{T}(1 + r_t)$
Sharpe ratio	$SR = \frac{R_{\text{ann}} - r_f}{\sigma_{\text{ann}}}, \qquad r_f = 0$
Maximum Drawdown	$MDD = \min_t \left(\frac{W_t}{\max_{s \le t} W_s} - 1 \right)$

Universe	Method	ann_return	ann_vol	sharpe	max_drawdown
Equity	Equal Weight	0.1755	0.1333	1.3171	-0.1039
Equity	Minimum Variance	0.0538	0.1097	0.4907	-0.0861
Equity	Maximum Sharpe	0.3911	0.1909	2.0486	-0.1216
Equity	Risk Parity	0.1422	0.1253	1.1351	-0.0963
Equity	Minimum CVaR	0.0885	0.1156	0.7656	-0.0721
Crypto	Equal Weight	0.9162	0.4982	1.8389	-0.2758
Crypto	Minimum Variance	1.5065	0.4182	3.6022	-0.1884
Crypto	Maximum Sharpe	0.9495	0.4605	2.0618	-0.3117
Crypto	Risk Parity	0.9652	0.4862	1.9851	-0.2680
Crypto	Minimum CVaR	1.5452	0.4367	3.5384	-0.1997
Combined	Equal Weight	0.2504	0.1515	1.6523	-0.1075
Combined	Minimum Variance	0.0538	0.1097	0.4907	-0.0861
Combined	Maximum Sharpe	0.2967	0.1770	1.6765	-0.1039
Combined	Risk Parity	0.1791	0.1300	1.3778	-0.0977
Combined	Minimum CVaR	0.0885	0.1156	0.7656	-0.0721

Table 1: Performance metrics table (Annualised return, volatility, sharp ratio, maximum drawdown)

Table 1 summarises the overall performance of all portfolio optimisation methods across the Equity, Cryptocurrency and Combined investment universes. Comparing multiple performance metrics provides a more complete assessment, as a higher return does not necessarily imply a better investment if it is accompanied by substantially higher risk.

Overall, the Cryptocurrency funds achieved the highest annualised returns and Sharpe ratios during the 2023 out-of-sample period, reflecting the strong recovery of the crypto market over the sample period. In comparison, the Equity funds delivered more moderate returns with lower volatility, while the Combined portfolios provided a balance between return and risk by diversifying across both asset classes. This highlights that portfolio performance is influenced not only by the optimisation method but also by the characteristics of the underlying assets.

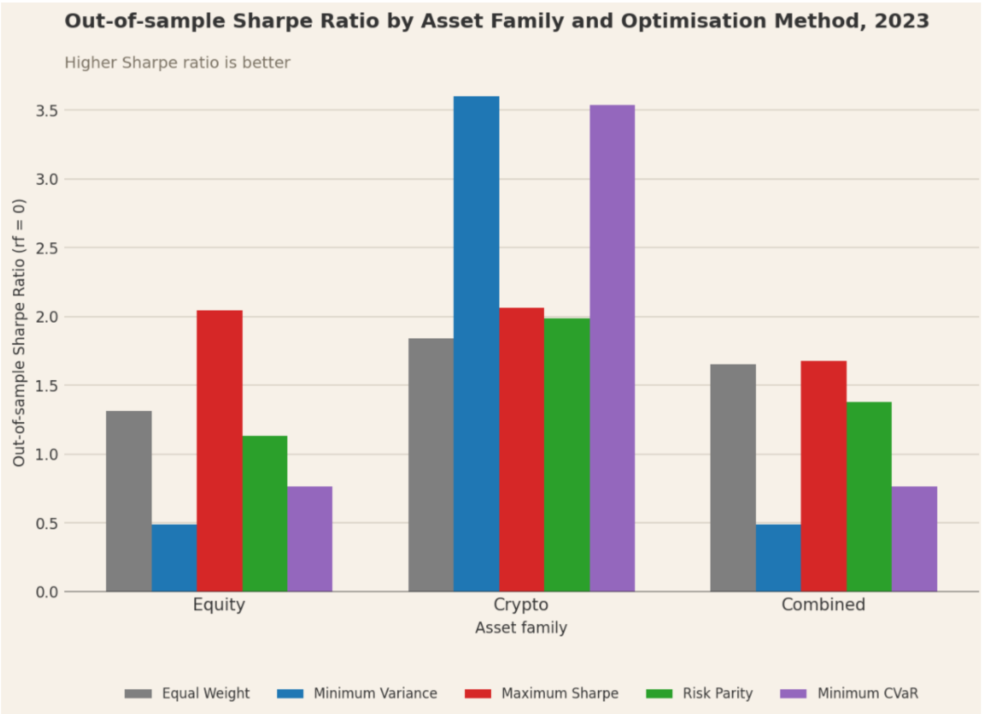

Figure 1: Out-of-sample Sharpe ratio by asset family and optimisation method

As shown in *Figure 1*, among the Equity funds, Maximum Sharpe achieved the highest Sharpe ratio, while the Cryptocurrency universe produced the strongest risk-adjusted performance overall. Most sharply for Minimum Variance and Minimum CVaR (both above 3.5). For Equal Weight, and Risk Parity, the Combined portfolios sit between their Equity and Crypto counterparts, consistent with genuine diversification, blending two asset classes traded some crypto upside for lower risk. Minimum Variance and Minimum CVaR tell a different story. Their Combined Sharpe ratios (0.49 and 0.76) are nearly identical to their Equity-only values, because both optimisers allocated close to zero weight to crypto (Section 4), avoiding its volatility almost entirely rather than diversifying into it. This shows the "diversification lowers risk" story applies unevenly. It holds for methods that actually hold both asset classes, not for those that optimise crypto out of the portfolio.

Figures 2 to 4 illustrate the cumulative growth of $1 invested in each optimisation method throughout the 2023 out-of-sample period. These figures show not only the final investment outcome but also how each portfolio performed over time under changing market conditions.

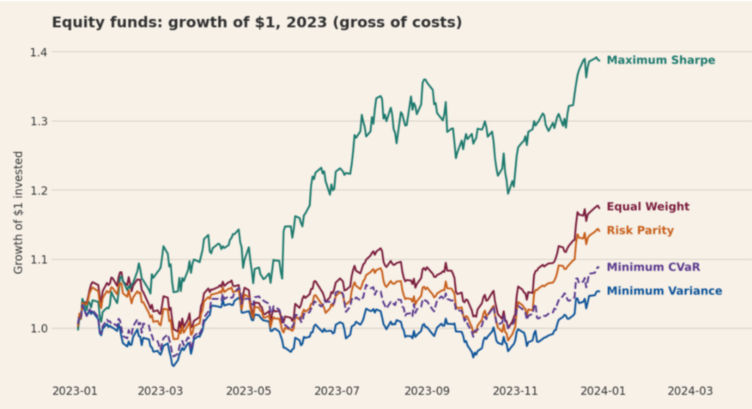

Figure 2: Equity Growth of $1 invested

For the Equity universe (*Figure 2*), Maximum Sharpe generated by far the highest cumulative return, reaching ~$1.4 by year-end, though visibly with the largest swings along the way. Minimum Variance and Minimum CVaR track closely together near the bottom of the range, consistent with their lower annualised volatility in *Table 1* (11.0% and 11.6%, the lowest of the five). Equal Weight and Risk Parity sit in between, ending around $1.15–1.18.

Figure 3: Crypto Growth of $1 invested

Figure 3 shows the Minimum Variance and Minimum CVaR portfolios delivering the strongest cumulative growth in the Cryptocurrency universe, reaching ~$2.5–2.6 by year-end versus ~$2.0 for the other three methods, consistent with *Table 1*, where both exceed 150% annualised return and a 3.5+ Sharpe ratio, against Maximum Sharpe's 94.9%/2.06. This is counter-intuitive. Maximum Sharpe was actually the weakest performer for most of the year (visible May–November) before a late recovery. A likely explanation is that crypto's high volatility makes mean returns hard to estimate reliably, by targeting risk stability instead of a noisy mean forecast, Minimum Variance and Minimum CVaR captured the 2023 uptrend without over-concentrating on an unstable estimate, a pattern also visible in their shallower drawdowns (see ***Appendix B, Figure 2)***.

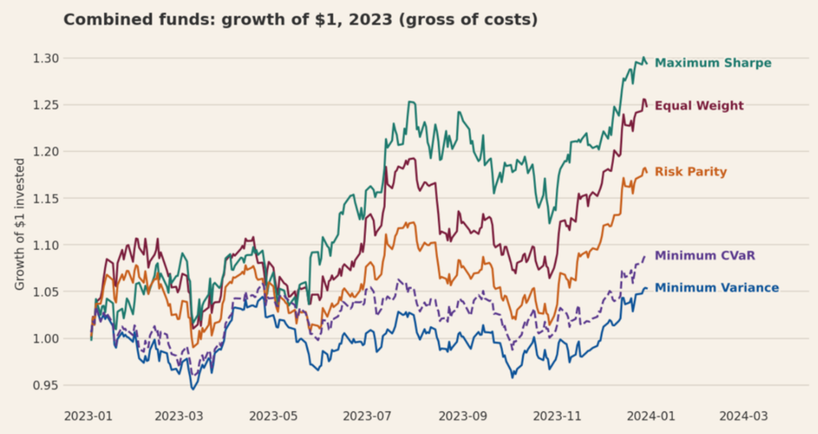

Figure 4: Combined Growth of $1 invested

The Combined portfolios show mixed results relative to their Equity-only counterparts (*Figure 4*). Equal Weight and Risk Parity achieved moderately higher returns from crypto exposure (25.0% and 17.9% vs. 17.6% and 14.2%), while Minimum Variance and Minimum CVaR were essentially unchanged, consistent with their near-zero crypto allocation discussed in Section 4. Maximum Sharpe is the exception, its Combined return (29.6%) was actually lower than Equity-only (39.0%), suggesting the optimiser's reallocation across the larger combined asset set did not preserve the concentrated equity positions that drove its standalone performance.

Although drawdown profiles for the Equity and Cryptocurrency portfolios are also presented in ***Appendix B, Figure 1 & 2,*** the Combined portfolio is discussed here because it best represents the diversified investment strategy designed for long-term retail investors. Drawdown measures the percentage decline from a previous portfolio peak and provides an indication of downside risk experienced before the portfolio recovers.

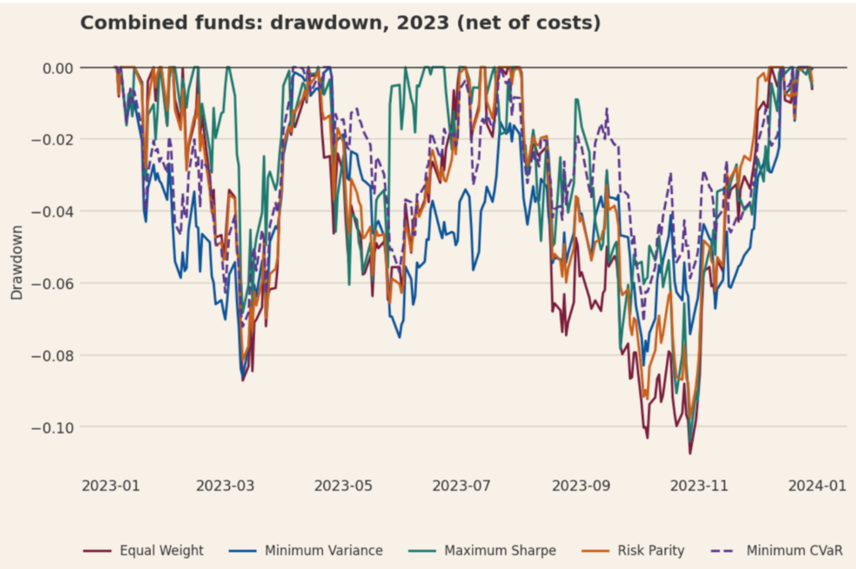

Figure 5: Combined portfolio drawdown

As shown in *Figure 5*, all portfolio strategies experienced a noticeable drawdown during October and November 2023, indicating that the decline was largely driven by broader market conditions rather than weaknesses in any individual optimisation method. This period coincided with heightened market uncertainty following the Israel-Hamas war's outbreak, though the dominant driver was likely the broader rates environment, markets were repricing for interest rates staying "higher for longer" through October, before the Fed's more dovish signal in December.

Despite experiencing the same market-wide shock, the Minimum Variance portfolio recovered more steadily during the October–November episode, consistent with its shallower drawdown profile throughout the sample. This comes at a real cost, however*,* *Table 1* shows Minimum Variance recorded the lowest annualised return (5.4%) and lowest Sharpe ratio (0.49) of all five Combined funds. For a retail investor, this represents a genuine trade-off rather than a "best of both" outcome. Minimum CVaR, however, delivers a shallower drawdown than Minimum Variance (−7.21% vs. −8.61%) alongside a higher return and Sharpe ratio in this sample. *Table 1* shows Minimum Variance recorded the lowest annualised return (5.4%) and Sharpe ratio (0.49) of all five Combined funds, a genuine trade-off for the stability it offers, and one an investor should weigh against the alternatives before choosing it.

Transaction costs were incorporated into all reported net performance metrics using a one-way cost of 5 basis points per rebalance. Although transaction costs slightly reduced portfolio returns, the differences between gross and net performance were minimal because the assumed trading cost was modest and portfolios were rebalanced only once per month. As shown in ***Appendix B, Figure 3***, the inclusion of transaction costs had little impact on the overall ranking of the baseline funds, indicating that the relative performance of the portfolio optimisation methods remained largely unchanged after implementation costs were considered.

Transaction costs:

$$\boldsymbol{TO_t = \tfrac{1}{2}\sum_i \left| w_{i,t} - w_{i,t-1} \right|,}$$

$$\boldsymbol{c_t = 0.0005\,TO_t, \qquad r_t^{net} = \left(1 + r_t^{gross}\right)(1 - c_t) - 1}$$

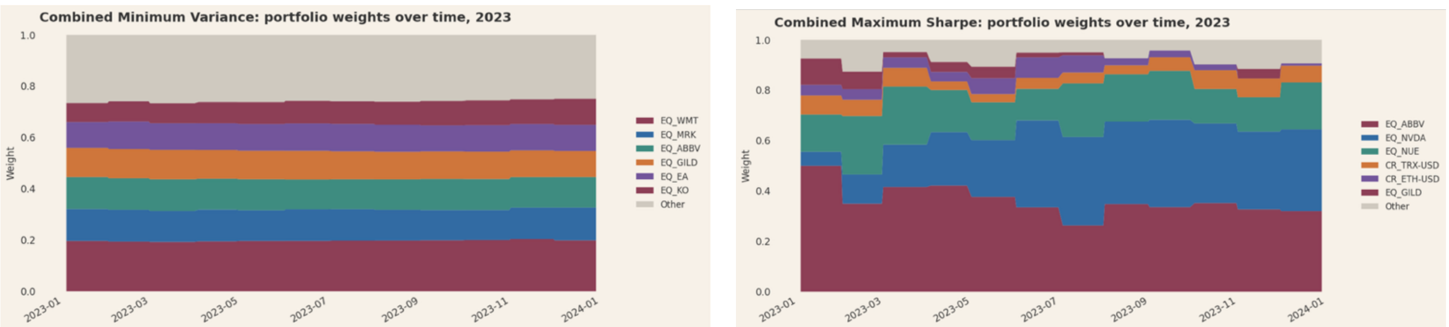

Figure 6: Combined portfolio weights over time — Minimum Variance vs Maximum Sharpe, 2023.
(Only the largest individual holdings are labelled, smaller positions are grouped as "Other" for readability.)

Figure 6 contrasts portfolio composition over the out-of-sample period for two optimisation methods applied to the same Combined fund. The Minimum Variance portfolio's allocation is remarkably stable throughout 2023, retaining broadly the same six holdings in similar proportions from January to December. This stability is mechanically reinforced by the expanding-window design, as each additional month contributes proportionally less to the estimated covariance structure, the covariance matrix becomes increasingly stable, and so do the resulting weights.

In contrast, the Maximum Sharpe portfolio reallocates substantially over the same period. GILD's weight falls from roughly 50% to 30% by March, while crypto led by TRX-USD, with ETH-USD and BTC-USD rotating in and out holds a fairly steady 12–17% throughout the year. This difference in turnover is not merely descriptive, it is the direct mechanical driver of the transaction cost impact reported in ***Appendix B, Figure 3***, where higher-turnover methods incur proportionally larger drag from the assumed 5bp per-rebalance cost.

3. The sentiment index

Having established baseline and CVaR-extended fund performance in Section 2, we next construct an independent sentiment signal from news headlines, which Section 4 tests as a possible enhancement to the Minimum Variance strategy.

The equity sentiment index was constructed using both VADER and finVADER to measure the sentiment contained in financial news headlines. While VADER is a general-purpose sentiment model, finVADER incorporates a finance-specific lexicon designed to better capture terminology commonly used in financial markets. To avoid look-ahead bias, all sentiment scores were lagged by one trading day before being used in subsequent portfolio construction.

Headline sentiment aggregation:

$$s_{i,t} = \frac{1}{H_{i,t}} \sum_{h=1}^{H_{i,t}} compound_h$$

Individual company sentiment scores were then aggregated into sector-level indices using an equal-weighted average across companies within each sector, so that no single company's news volume dominates the sector signal. A coverage-weighted alternative, where companies with more headlines receive proportionally greater influence was also tested as a robustness check **(**see ***Appendix B, Figure 4*****).** The two measures track closely for most sectors and diverge only for Utilities in early 2020, when coverage across all three thin-coverage sectors was sparsest. Equal weighting was retained as the primary index because it avoids letting the most heavily covered, typically larger, more newsworthy. Companies dominate a sector's sentiment reading, and the comparison shows this simpler approach is not materially different from coverage-weighting once headline coverage stabilises. A 21-day moving average was subsequently applied to smooth short-term fluctuations and better capture underlying sentiment trends.

Sector sentiment index:

$$S_{k,t} = \frac{1}{N_k} \sum_{i \in k} s_{i,t}$$

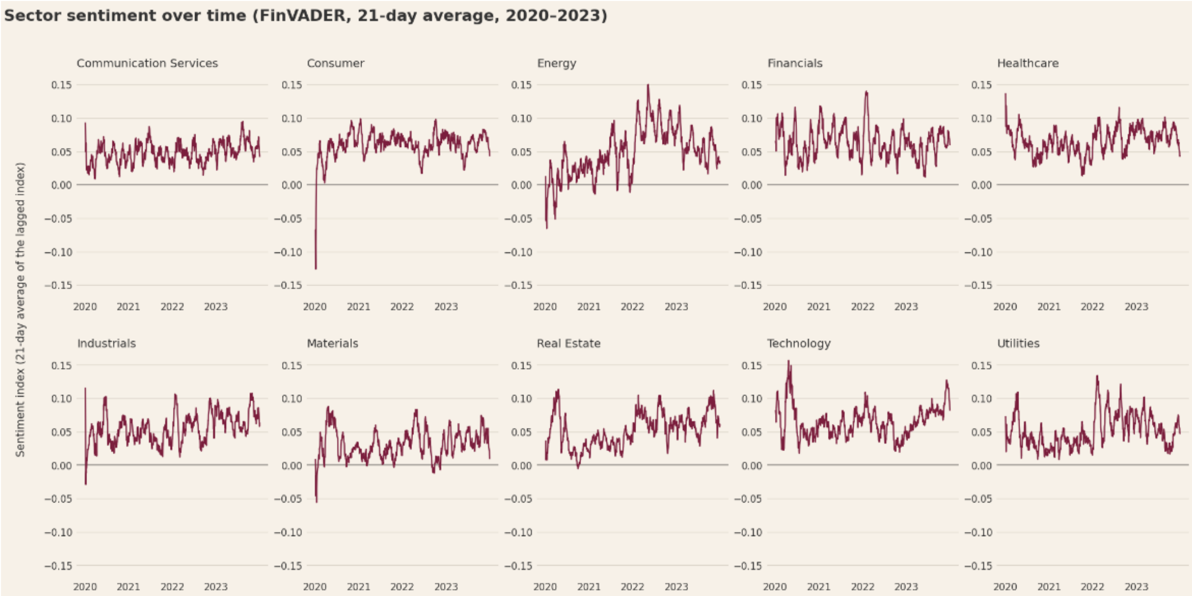

Figure 7: Sector sentiment over time (finVADER)

Figure 7 presents the lagged finVADER sector sentiment indices from 2020 to 2023. Overall, sentiment remained above zero for most sectors throughout the sample period, suggesting that the overall tone of financial news was generally positive. A noticeable decline occurred across almost all sectors during the COVID-19 market disruption in early 2020, after which sentiment gradually recovered as market conditions improved.

Although all sectors responded to major market events, each sector exhibited distinct sentiment dynamics over time. For example, the Energy sector experienced a sustained increase in sentiment during 2021–2022, while the Technology sector showed a gradual upward trend towards the end of the sample period. In contrast, Consumer and Financials remained relatively stable following the initial COVID-19 disruption. These differences suggest that the sentiment index captures both market-wide events and sector-specific information rather than simply reflecting overall market movements.

The relatively smooth trends observed across sectors also reflect the use of coverage weighting together with the 21-day moving average. By giving greater influence on companies receiving more news coverage while reducing the impact of isolated company-specific headlines, the resulting sector sentiment index provides a more stable signal for subsequent portfolio construction.

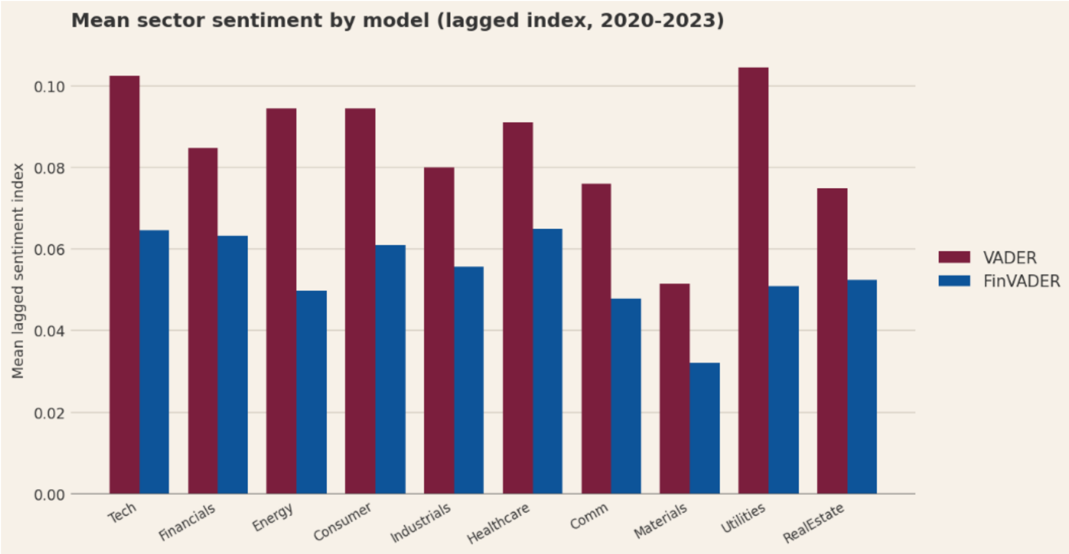

Figure 8: Comparison between VADER and finVADER

Figure 8 compares the average lagged sector sentiment generated by VADER and finVADER across all equity sectors. Both models produced broadly similar sector rankings, indicating that they captured consistent patterns in sentiment across industries. However, VADER consistently generated higher average sentiment scores than finVADER across all sectors.

One plausible explanation is that VADER's general-purpose lexicon interprets financial headlines more positively because business news frequently employs optimistic language, words such as "growth," "expansion," and "record" carry positive valence in VADER's general dictionary. FinVADER's finance-specific lexicon, by contrast, may correctly identify more nuanced or cautious financial terminology that tempers the overall score.

The magnitude of these differences also varied between sectors. Larger gap were observed in sectors such as Utilities, Technology and Consumer, whereas Healthcare and Financials exhibited comparatively smaller differences. This suggests that incorporating a finance-specific lexicon changes the strength of the measured sentiment while largely preserving the relative ordering of sectors.

Because the two sentiment models produce different sentiment signals, both were retained for the subsequent portfolio evaluation. Differences in the sentiment index alone do not indicate which model leads to better investment performance. Instead, their practical value is assessed in the following section by incorporating each sentiment index into the portfolio optimisation process and comparing the resulting out-of-sample performance.

4. The fusion extension and whether it adds value

Building on the sentiment index constructed in Section 3, this section evaluates whether it adds economic value. To evaluate whether incorporating news sentiment adds value to portfolio performance, this section focuses on the Equity Minimum Variance portfolio. The Minimum Variance strategy was selected because it best aligns with the needs of long-term retail investors by prioritising capital preservation and lower portfolio volatility. Consequently, only two of the four nominal fusion combinations are economically distinct. Equity Min-Variance +

VADER/finVADER, and Combined Min-Variance + VADER/finVADER, which because the Combined baseline holds effectively 0% crypto replicate the Equity results almost exactly rather than constituting an independent test.

Rather than constructing a new portfolio, news sentiment is incorporated through a sentiment tilt, which adjusts the baseline portfolio by increasing the weights of stocks with relatively positive news sentiment and reducing the weights of stocks with relatively negative sentiment while maintaining a fully invested long-only portfolio. This allows the impact of both VADER and finVADER sentiment models to be evaluated relative to the original Minimum Variance strategy.

Standardise Sentiment:

$$z_{i,t} = \frac{s_{i,t} - \bar{s}_t}{\sigma_{s,t}},$$

Apply the tilt:

$$\tilde{w}_{i,t} = max\left(0, w_{i,t}^{base}(1 + \lambda z_{i,t})\right),$$

Renormalised:

$$w_{i,t}^{fused} = \frac{\tilde{w}_{i,t}}{\sum_j \tilde{w}_{j,t}}$$

Note:

$\lambda = 0.5$ *was predetermined rather than selected using 2023 results.*

The resulting weight changes are shown below:

$$\overline{\Delta w_i} = \frac{1}{T}\sum_{t=1}^{T}\left(w_{i,t}^{fused} - w_{i,t}^{base}\right)$$

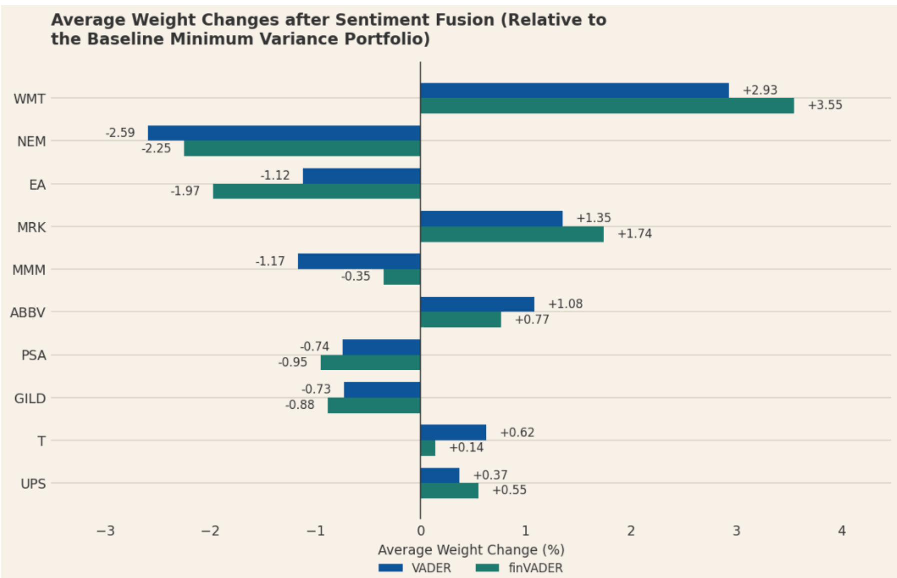

Figure 9: Average weight changes after sentiment fusion

As shown in *Figure* 9, both VADER and finVADER produced broadly similar portfolio tilts. Both models consistently increased allocations to Walmart (WMT) and Merck (MRK), while reducing exposure to Newmont (NEM) and Electronic Arts (EA). The similarity in the direction of these adjustments suggests that the two sentiment models generally agreed on which stocks should be overweighted or underweighted.

However, the magnitude of the tilts differed slightly. For example, finVADER assigned a larger overweight to WMT (+3.55%) than VADER (+2.93%), whereas VADER placed greater emphasis on ABBV and T. These differences in portfolio weights ultimately led to different investment outcomes evaluated in the following figures.

Portfolio	Ann.Return	Sharpe	Max Drawdown	Turnover
Baseline (Minimum Variance)	5.38%	0.490	-8.61%	0.05%
+ VADER	7.32%	0.662	-8.24%	0.95%
+ finVADER	4.97%	0.453	-8.80%	0.66%

Table 2: Performance of baseline and sentiment-enhanced Equity Minimum Variance portfolios

Table 2 summarises the out-of-sample performance of the baseline Equity Minimum Variance portfolio and its VADER- and finVADER-enhanced alternatives. Applying the VADER sentiment tilt increased annualised return from 5.38% to 7.32% and improved the Sharpe ratio from 0.490 to 0.662. Maximum drawdown also improved slightly, from −8.61% to −8.24%. These results indicate that the VADER-enhanced portfolio generated higher returns per unit of risk while providing marginally better downside protection during the evaluation period.

However, this improvement required greater trading activity. Turnover increased from 0.05% for the baseline portfolio to 0.95% following the VADER tilt. The economic value of the strategy therefore depends on whether its additional risk-adjusted return remains after transaction costs, which is examined in the subsequent analysis.

In contrast, finVADER reduced annualised return to 4.97% and lowered the Sharpe ratio to 0.453, while maximum drawdown deteriorated slightly to −8.80%. It also increased turnover to 0.66%. Therefore, finVADER generated additional portfolio reallocation without a corresponding improvement in return or risk. This finding demonstrates that greater financial-language specialisation does not necessarily produce a more valuable investment signal. The effectiveness of sentiment must ultimately be assessed through realised portfolio outcomes.

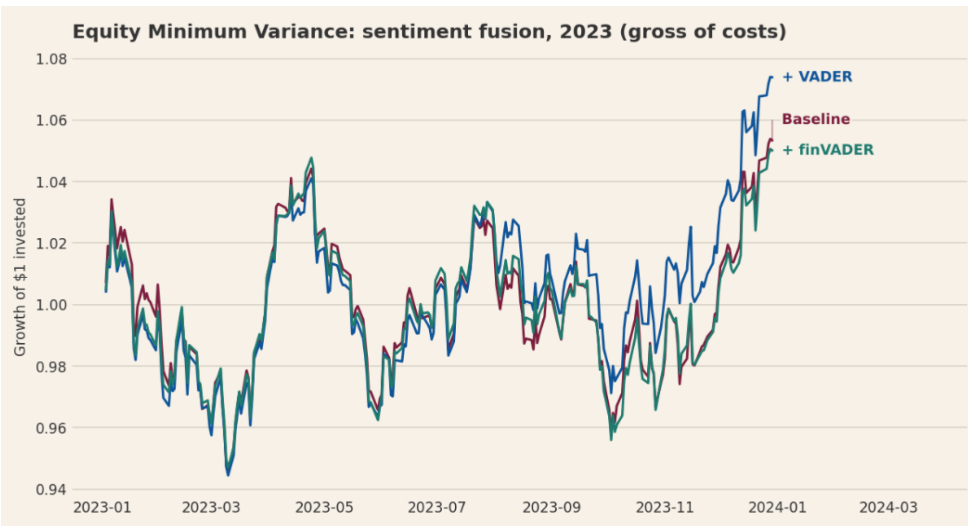

Figure 10: Growth of $1 invested after sentiment fusion

Figure 10 compares the cumulative out-of-sample performance of the baseline Minimum Variance portfolio with portfolios incorporating VADER and finVADER sentiment.

Although both sentiment models generated similar portfolio tilts, they produced noticeably different investment outcomes. The VADER-enhanced portfolio achieved the highest cumulative return by the end of the sample period, whereas the finVADER portfolio closely tracked the baseline and slightly underperformed towards the end of the evaluation period.

This result demonstrates that producing similar weight adjustments does not necessarily lead to similar investment performance. Even relatively small differences in portfolio weights can have a meaningful impact once compounded through time. In this case, the stocks receiving larger allocations under the VADER model subsequently generated stronger realised returns than those emphasised by finVADER, and this return differential compounded over the twelve-month out-of-sample window to produce a material separation in cumulative wealth.

Although finVADER was specifically designed for financial language, its specialised financial vocabulary did not translate into superior portfolio performance. This highlights that the effectiveness of a sentiment model should ultimately be evaluated using realised portfolio performance rather than the sophistication of the underlying sentiment classifier.

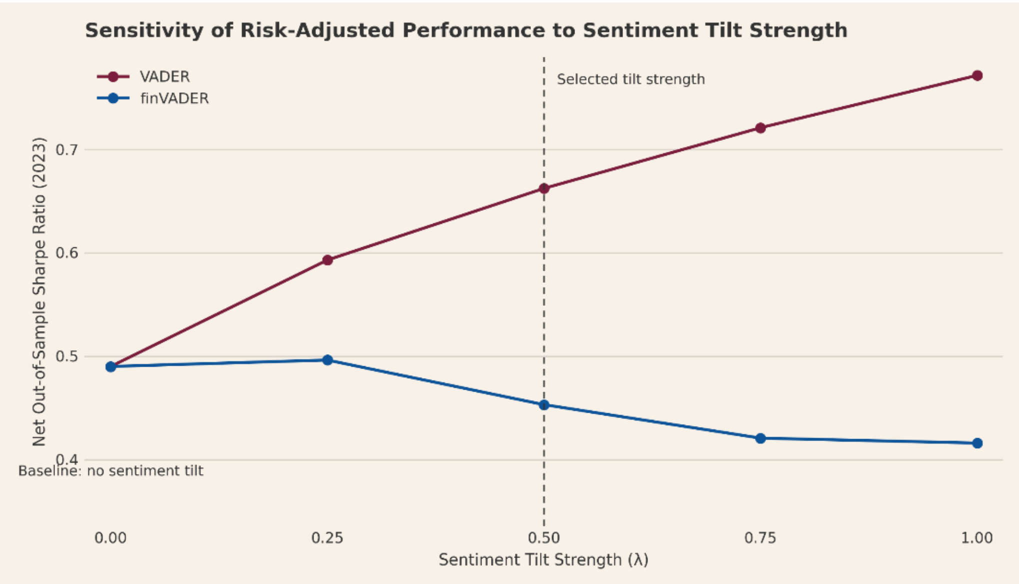

Figure 11: Sensitivity of net out-of-sample Sharpe ratios to sentiment tilt strength. *At λ =0, no sentiment tilt is applied. The dashed line identifies the predetermined value of λ =0.5 used in the main fusion analysis.*

Figure 11 examines whether the performance of the sentiment-enhanced portfolios is sensitive to the strength of the portfolio tilt, represented by λ .At λ = 0, no sentiment adjustment is applied, so both models reproduce the baseline Minimum Variance portfolio with a Sharpe ratio of approximately 0.49. As the strength of the VADER tilt increased, its net out-of-sample Sharpe ratio improved consistently, reaching approximately 0.66 at the selected value of λ =0.5. In contrast, finVADER produced only a marginal improvement at λ =0.25, after which its performance deteriorated as the portfolio placed greater weight on its signal.

The widening difference indicates that the economic value of sentiment depends on both the sentiment model and the strength with which it is incorporated into portfolio weights. During the 2023 out-of-sample period, stronger reliance on VADER improved risk-adjusted performance, whereas stronger reliance on finVADER amplified a less effective investment signal. This supports the selection of VADER for the news-aware portfolio. However, it does not establish that increasingly aggressive VADER tilts will perform better in future periods. The results are based on a single out-of-sample year, and selecting the highest-performing value after observing the evaluation results would introduce data-snooping bias.

The ranking of the three portfolios remained unchanged after incorporating trading costs (see ***Appendix B, Figure 5)***, indicating that the assumed transaction cost of 5 basis points per rebalance had only a limited effect on performance. The VADER-enhanced portfolio continued to achieve the highest Sharpe ratio, suggesting that its improvement was not solely the result of ignoring implementation costs. From an investment perspective, this indicates that the additional return generated by the VADER signal was sufficient to offset the modest increase in trading costs under the assumptions used in this study.

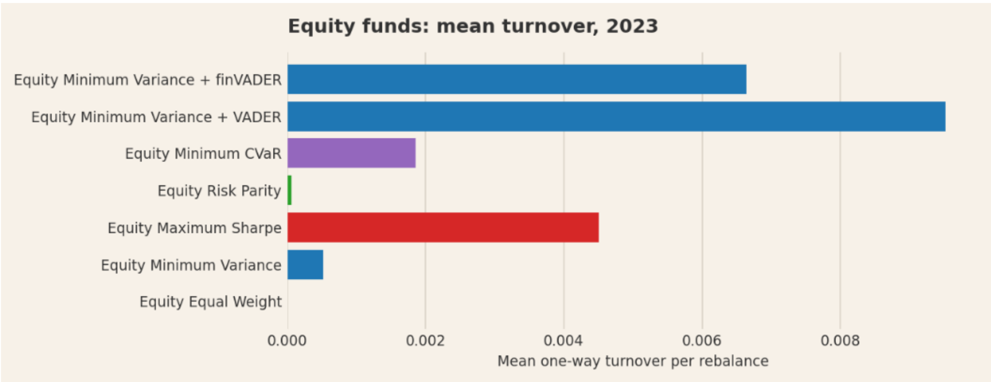

Figure 12: Portfolio turnover after sentiment fusion

Figure 12 shows that the sentiment tilts required substantially more portfolio rebalancing than the baseline Minimum Variance strategy. Because portfolio weights were adjusted whenever the sentiment signal changed, both sentiment-enhanced portfolios traded considerably more frequently, with VADER generating the highest turnover.

Higher turnover generally leads to greater transaction costs and may reduce the practical attractiveness of active portfolio management. However, despite the additional trading activity, the VADER portfolio continued to achieve the highest Sharpe ratio after costs, indicating that the performance gains were sufficient to compensate for the higher implementation costs. By contrast, finVADER also increased turnover but failed to improve risk-adjusted returns. This suggests that increasing trading activity alone does not create value. The quality of the underlying trading signal remains the key determinant of investment performance.

5. The app and the investor journey.

EverVest is designed for a **long-term retail investor** with limited quantitative finance background. Someone who wants systematic, diversified exposure to equities and crypto but lacks the tools or expertise to build and backtest portfolios themselves. Thesw users value transparency and plain-language explanation over technical sophistication. They prefer comparing a handful of valuable investment options rather than actively trading and needs to understand risk (not just return) before committing money.

1. Home Page: Choosing an Investment Objective

The investor journey begins on the Home page, where users explore funds according to their investment objectives rather than unfamiliar optimisation terminology. Plain English naming is designed for investors to understand each method easily, such as Minimum Variance as “Capital Preservation”, Minimum CVaR as “Downside Protection”, Maximum Sharpe as “Risk-Adjusted Growth” etc. Each objective also includes a brief plain-language explanation and is available across the Equity, Crypto and Diversified universes. The technical method remains visible beneath its investor-friendly name, maintaining methodological transparency while reducing the knowledge required to begin exploring the funds.

The Home page also distinguishes the Equity News-Aware Capital Preservation fund as a research-enhanced strategy. This fund combines the Equity Minimum Variance portfolio with the VADER sentiment tilt. It is presented separately because it represents the application of the fusion experiment rather than another baseline optimisation method. Users can access its relevant fact sheet directly from the strategy card.

2. Compare Funds: Evaluating the Alternatives

After selecting an objective, users can proceed to the Compare Funds page. This page supports comparison across investment universes, objectives and fund types. Users can filter the available strategies and assess their annualised return, volatility, Sharpe ratio and maximum drawdown on either a gross or net-of-cost basis anytime.

Organising the funds by investment objective helps users understand why each strategy exists. It also avoids presenting the available portfolios as a long list of unrelated technical methods. The comparison process encourages users to consider risk-adjusted performance and downside risk rather than ranking funds solely by return.

3. Fund Fact Sheet: Understanding an Individual Fund

After identifying a potential strategy, users can open its Fund Fact Sheet. This page provides the selected fund’s key performance measures, plain-language objective, technical methodology, current holdings and historical growth of $1.

The fact sheet connects the fund’s stated objective with its realised out-of-sample behaviour. For example, a user considering a Capital Preservation fund can examine not only its return but also its volatility and maximum drawdown. Combining quantitative evidence with plain-language explanations allows users to understand both the potential benefit and risk of each fund.

4. Allocate: Building a Portfolio

The Allocate page completes the principal investment journey by allowing users to combine up to five funds and assign a percentage allocation to each. The application verifies that the selected weights total 100% and calculates the blended portfolio's annualised return, volatility, Sharpe ratio and maximum drawdown using the funds' precomputed out-of-sample returns.

This feature transforms passive fund comparison into an interactive allocation decision. It allows users to decide and observe how combining different strategies changes overall portfolio behaviour and demonstrates that portfolio construction involves considering the relationship between funds rather than selecting only the individually highest-returning strategy.

The results remain historical illustrations rather than forecasts. The allocation tool does not currently assess personal circumstances or determine whether a portfolio is suitable for an individual investor.

5. Market Insights: Understanding the Sentiment Signal

The Market Insights page presents the unstructured-data component of Project B without treating sentiment as a standalone investment recommendation. It explains the historical tone of company news, compares sentiment across sectors and demonstrates how news scores affect portfolio weights through the sentiment tilt.

The page also presents the evidence supporting the selection of VADER for the News-Aware Capital Preservation strategy. More technical comparisons, including VADER versus finVADER headline scores, the sector sentiment index and tilt-strength sensitivity are placed inside expandable research sections. This preserves the analytical evidence while preventing technical detail from overwhelming the main investor journey.

The page clearly states that sentiment measures the tone of historical financial news and does not predict returns with certainty.

6. Methodology: Making the Process Transparent

The Methodology page documents the calculations underlying the application, including return construction, the expanding-window backtest, portfolio optimisation, performance measures, transaction costs, sentiment aggregation and the fusion mechanism. Separating this material from the main investor journey allows interested users and assessors to inspect the technical process without forcing all users to interpret equations before comparing funds. It also demonstrates that the figures displayed in the application are produced through a documented and reproducible methodology.

7. About the Data: Assumptions and Limitations

The About the Data page explains the investment universes, study period, rebalancing schedule, annualisation conventions, transaction-cost assumption and limitations of the analysis. It clarifies that the strategies were estimated using the 2020–2022 in-sample period and evaluated using unseen data from 2023. This page also communicates the principal limitations of the product. In particular, one out-of-sample year provides only limited evidence of long-term strategy performance, headline sentiment is noisy, and the assumed transaction costs simplify real implementation conditions.

Overall, the investor journey follows a clear sequence:

Choose an objective → Compare funds → Read a fact sheet → Construct an allocation → Explore supporting market insights → Review the methodology and limitations.

6. Critical reflection with three concrete recommendations.

The project successfully converted portfolio research into an investor-facing application, but the results also revealed limitations that would need to be addressed before EverVest could operate as a real investment platform. In particular, the short evaluation period limits confidence in the performance rankings, sentiment added value only under specific conditions, and the allocation tool currently provides analysis without assessing investor suitability.

Recommendation 1: Validate the funds across multiple market conditions

The expanding-window backtest reduced look-ahead bias and produced meaningful differences between funds. However, the entire out-of-sample evaluation covers only 2023. This is not sufficient to conclude that the highest-performing funds possess a persistent advantage. The year may have favoured assets or strategies, especially those exposed to recovering equity and cryptocurrency markets. A fund with the highest 2023 Sharpe ratio could perform differently during a prolonged market decline, inflation shock or liquidity crisis. The 2023 crypto recovery, particularly followed the severe 2022 drawdown associated with the FTX collapse, testing only the rebound period risks overestimating the attractiveness of crypto-heavy strategies while underestimating their downside potential in a sustained bear market.

Before offering the funds to real investors, EverVest should extend the walk-forward test across at least one complete market cycle and report results separately for rising, falling and high-volatility markets. Fund rankings should be based on consistency across periods, not the highest return in one year. The app could communicate this uncertainty by displaying a "market-regime coverage" indicator alongside each fund's historical metrics, together with confidence intervals or worst-case drawdown estimates from the full test history.

Recommendation 2: Deploy sentiment only as a controlled secondary signal, with model-specific limits

VADER improved the Equity Minimum Variance portfolio's annualised return from 5.38% to 7.32% and its Sharpe ratio from 0.490 to 0.662, even after transaction costs. However, finVADER reduced annualised return to 4.97%, lowered the Sharpe ratio to 0.453.

The tilt-sensitivity analysis reinforced this difference. Stronger VADER tilts improved the 2023 Sharpe ratio, whereas stronger finVADER tilts generally weakened it. These results demonstrate that a model's greater linguistic specialisation does not guarantee a more useful investment signal. The finance-specific vocabulary may have generated stronger reactions to information that was already reflected in prices, or the fixed fusion rule may have responded too strongly to noisy headlines.

In a real implementation, EverVest should therefore retain the underlying Minimum Variance strategy as the portfolio's core and use VADER only as a controlled secondary adjustment. The sentiment tilt should be subject to a maximum change per asset, a minimum signal threshold and a no-trade region for small changes.

The tilt strength must be selected using training data or nested walk-forward validation rather than the final evaluation period. EverVest should also monitor whether the sentiment-enhanced fund continues to outperform its unfused baseline after costs. If that advantage disappears over a predefined review period, the tilt should be reduced or suspended. finVADER should remain a research model until it demonstrates economic value on genuinely unseen data.

Recommendation 3: Add investor-suitability and portfolio-risk controls to the allocation journey

The Compare Funds and Fact Sheet pages allow users to examine risk-adjusted performance and holdings, while the Allocate page turns this information into an interactive portfolio decision. However, the current allocation tool allows users to combine funds and observe historical portfolio metrics without determining whether the allocation is appropriate for them. Two differently named funds may also hold overlapping assets, meaning that selecting several products does not necessarily provide the diversification the user expects. Historical volatility and drawdown alone cannot determine suitability, particularly when cryptocurrency exposure is involved.

A real-world version of EverVest should introduce a short risk-and-horizon assessment before allocation. It should ask about investment horizon, loss tolerance, liquidity needs and willingness to hold cryptocurrency. The resulting allocation page should show total equity and cryptocurrency exposure, overlapping holdings, concentration and each fund's contribution to portfolio risk. It should also warn users when an allocation conflicts with their stated preferences—for example, when a conservative investor assigns a large proportion to a volatile crypto strategy. These controls should guide education and comparison rather than claim to provide personalised financial advice unless the platform satisfies the relevant regulatory requirements.

Overall, the project worked best when it combined disciplined portfolio construction with transparent communication. The expanding-window backtest improved methodological credibility, VADER demonstrated that sentiment could add value under a controlled fusion design, and the app translated technical strategies into a coherent investor journey. What did not work was equally informative, finVADER's specialised lexicon did not improve investment outcomes, higher turnover did not always create economic value, and one out-of-sample year could not establish long-term robustness. The three recommendations therefore focus on longer validation, controlled use of sentiment and stronger investor-protection mechanisms, the principal requirements for moving EverVest from a coursework prototype towards a credible real-world investment platform.

Appendix

Appendix A: Methodology

Minimum CvaR:

$$\sum_{i=1}^{N} w_i = 1, \qquad w_i \geq 0$$

$$z_t \geq -\mathbf{w}^\top \mathbf{R}_t - \beta, \qquad z_t \geq 0, \qquad \alpha = 0.95$$

where:

- β represents the portfolio's Value at Risk threshold
- z_t captures losses exceeding that threshold
- $\mathbf{R}_t$ is the vector of asset returns in historical scenario \(t\)
- T is the number of training observations
- $\alpha = 0.95$ means the model minimises expected losses within the worst 5% of outcomes
- $r_f = 0$ in your Maximum Sharpe implementation.

Annualised return and volatility:

$$R_{\text{ann}} = \left[\prod_{t=1}^{T}\left(1 + r_{p,t}\right)\right]^{A/T} - 1$$

Annualised volatility is: $\sigma_{\text{ann}} = \sigma\left(r_p\right)\sqrt{A}$

Where:

$$A = \begin{cases} 252, & \text{Equity and Combined portfolios} \\ 365, & \text{Cryptocurrency portfolios} \end{cases}$$

Daily asset return:

$$r_{i,t} = \frac{P_{i,t}^{adj}}{P_{i,t-1}^{adj}} - 1$$

where $P_{i,t}^{adj}$ is the adjusted closing price of asset i on day t.

Twenty-one-day sentiment average:

$$\overline{S}_{k,t} = \frac{1}{21}\sum_{j=0}^{20} S_{k,t-j}^{lag}$$

Coverage-weighted sector sentiment:

$$S_{k,t}^{cov} = \frac{\sum_{i \in k} n_{i,t}\, s_{i,t}}{\sum_{i \in k} n_{i,t}}$$

where:

- k is the sector
- $n_{i,t}$ is the number of headlines for company \(i\)
- $s_{i,t}$ is its ticker-day sentiment.

Appendix B: Figures

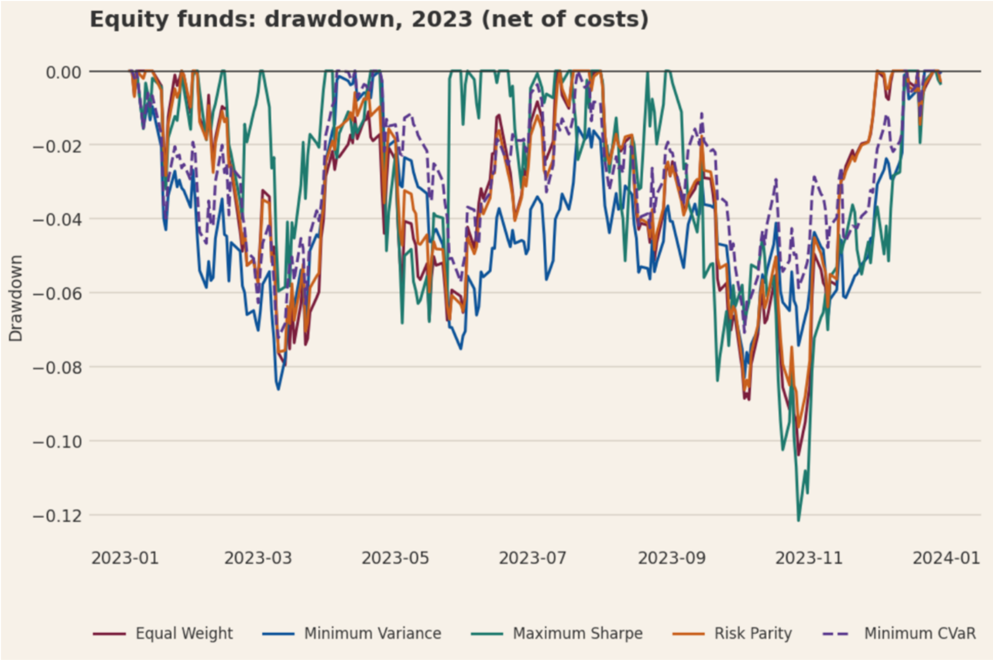

Figure 1: Equity portfolio drawdown

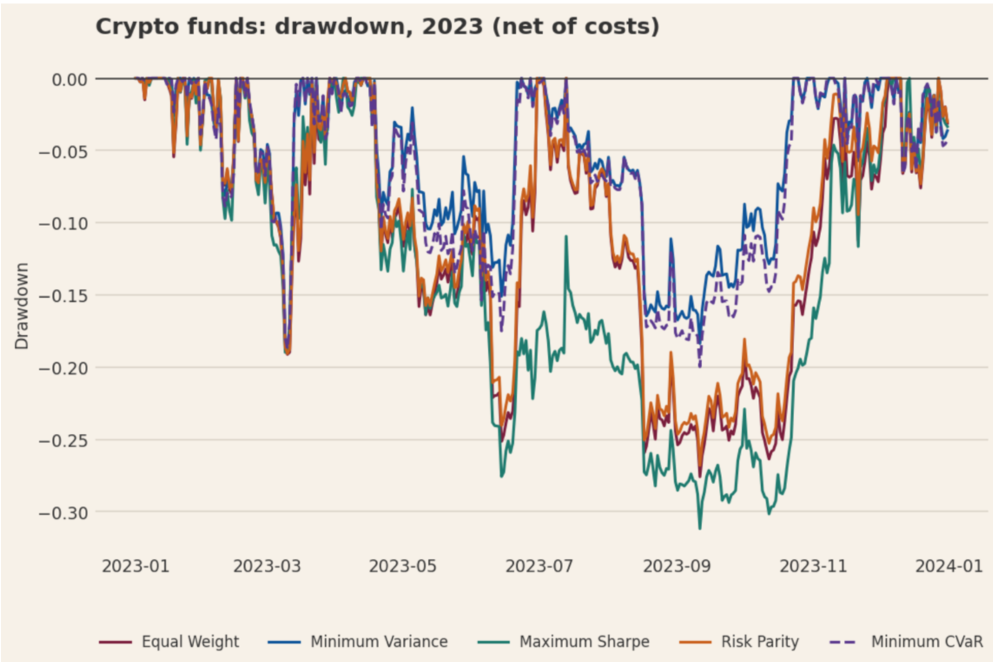

Figure 2: Crypto portfolio drawdown

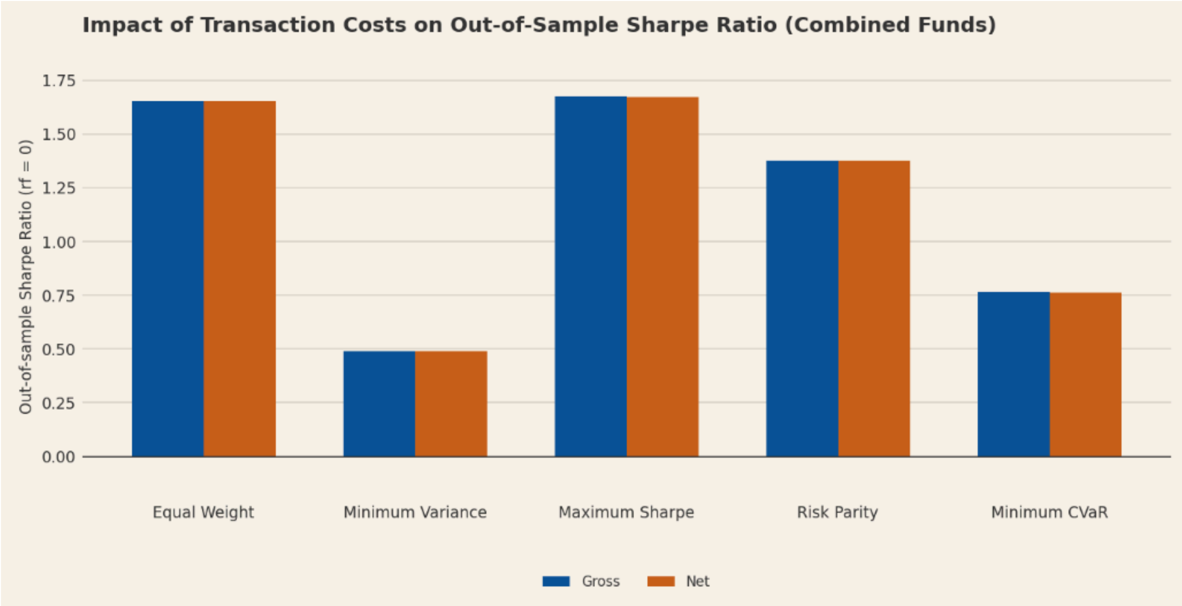

Figure 3: Performance after transaction costs

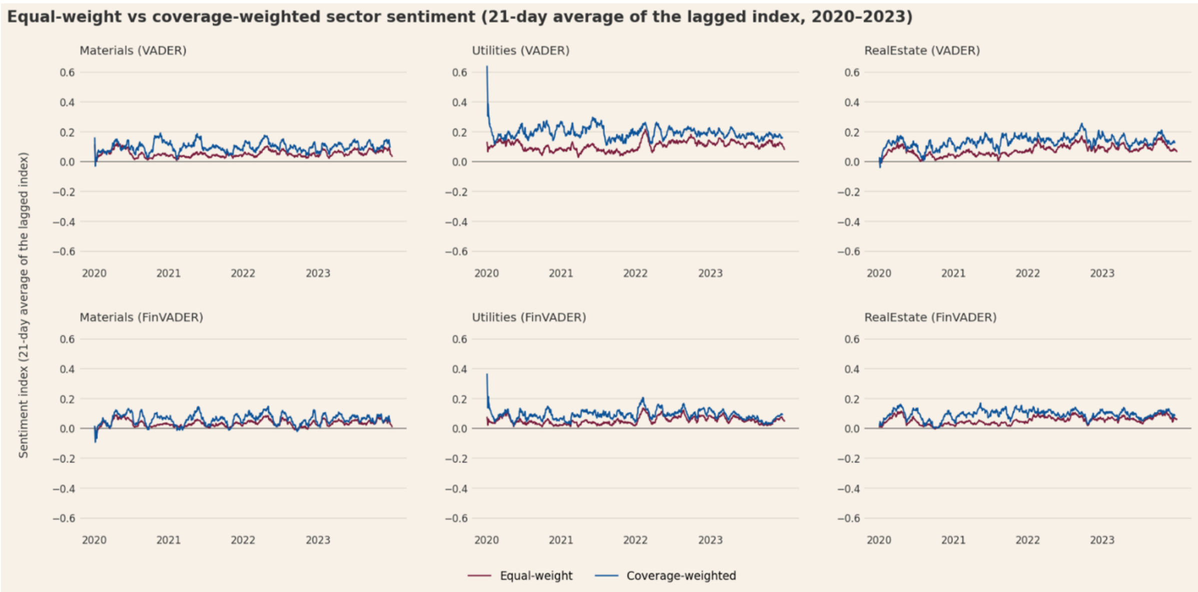

Figure 4: Equal-weight vs. coverage-weighted sector sentiment, 2020–2023

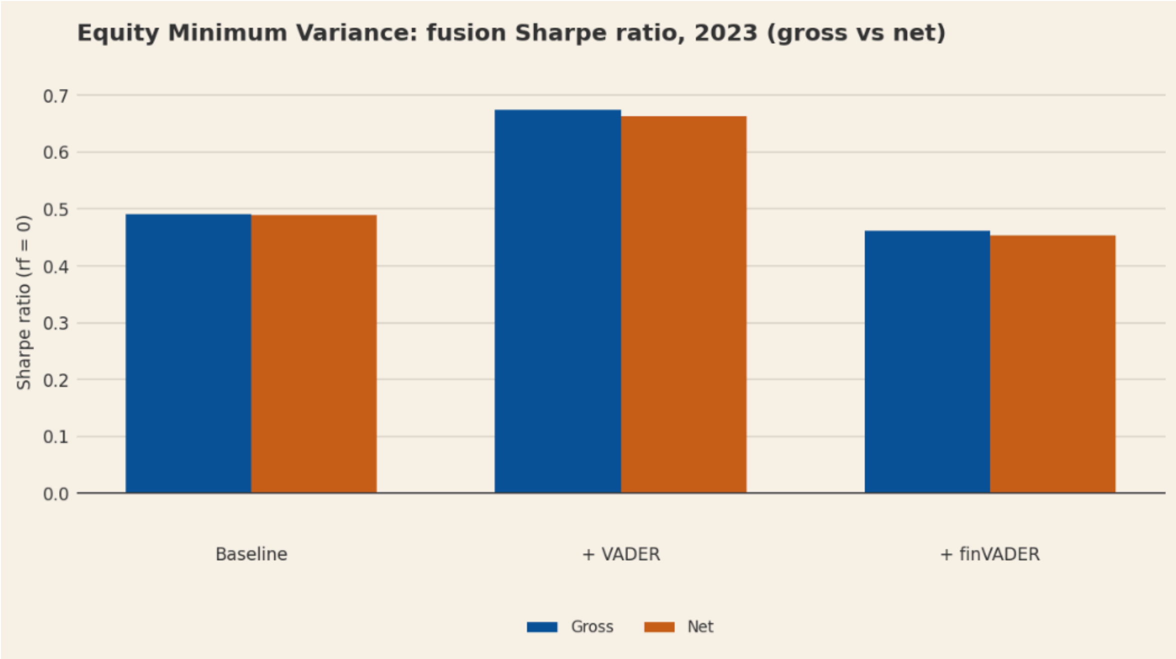

Figure 5: Equity Minimum Variance — fusion Sharpe ratio, 2023 (gross vs net)